Pattern Recognition Assignment#1

November 14, 2023

Q1. The maximum likelihood estimate is not always an unbiased estimator. Prove that the maximum likelihood estimator of the variance of a Gaussian variable is biased.

A1. **Proof 1** *Consider a Gaussian random variable, for which the density function of X is parameterized by $\theta = (\mu, \sigma)$ and is defined as follows:*

$$\mathcal{P}_\theta(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

We obtain the maximum likelihood estimates:

$$\hat{\mu} = \frac{1}{m}\sum_{i=1}^{m} x_i \quad \text{and} \quad \hat{\sigma} = \sqrt{\frac{1}{m}\sum_{i=1}^{m}(x_i - \hat{\mu})^2}.$$

We remark that the proof holds for every random variable with finite variance. Let $\mu = \mathbb{E}[x_1] = \ldots = \mathbb{E}[x_m]$ and let $\mu' = \mathbb{E}[x_1^2] = \ldots = \mathbb{E}[x_m^2]$. Note that for $i \neq j, E[x_i x_j] = \mathbb{E}[x_i]\,\mathbb{E}[x_j] = \mu^2$.

$$\begin{aligned}
(\mathbb{E}[\hat{\sigma}])^2 &= \frac{1}{m}\sum_{i=1}^{m}\left(\mathbb{E}\left[x_i^2\right] - \frac{2}{m}\sum_{j=1}^{m}\mathbb{E}[x_i x_j] + \frac{1}{m^2}\sum_{j,k}\mathbb{E}[x_j x_k]\right) \\
&= \frac{1}{m}\sum_{i=1}^{m}\left(\mu' - \frac{2}{m}\left((m-1)\mu^2 + \mu'\right) + \frac{1}{m^2}\left(m\mu' + m(m-1)\mu^2\right)\right) \\
&= \frac{1}{m}\sum_{i=1}^{m}\left(\frac{m-1}{m}\mu' - \frac{m-1}{m}\mu^2\right) \\
&= \frac{1}{m}\frac{m(m-1)}{m}\left(\mu' - \mu^2\right) \\
&= \frac{m-1}{m}\sigma^2.
\end{aligned}$$

Hence, the estimate $\hat{\sigma}$ of the variance is biased.

Q2. Consider the Bayesian belief network as shown in Figure 1.

(a) Which of the following are asserted by the network structure and why?

i. $P(B, I, L) = P(B)P(I)P(L)$.

ii. $P(J|G) = P(J|G, I)$.

iii. $P(L|G, B, I) = P(L|G, B, I, J)$.

(b) Calculate the value of $P(B = t, I = t, L = f, G = t, J = t)$.

(c) Calculate the probability that someone will go to jail given that he/she has broken the law, has been indicted, and his/her lawyer is inexperienced.

(d) Suppose we want to add the variable P = Presidential Pardon to the network, draw the new network and briefly explain any links you add.

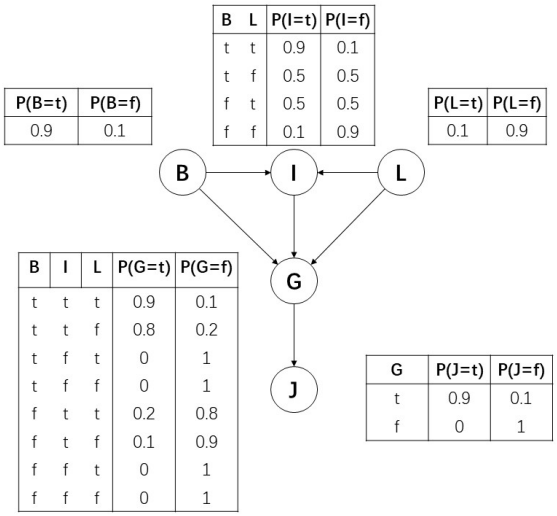


Figure 1: A Bayesian belief network with Bernoulli-distributed variables B = Broke Law, I = Indicted, L = Inexperienced Lawyer, G = Found Guilty, J = Jailed.

A2. (a) Directed connections between nodes represent dependencies between variables.

i. It doesn't hold because $P(B, I, L) = P(I|B, L)P(B)P(L)$ and clearly $P(I|B, L) \neq P(I)$.

ii. It holds because there is no connection between J and I.

iii. It holds because there is no connection between L and J.

(b)

$$
\begin{aligned}
&P(B = t, I = t, L = f, G = t, J = t) \\
&= P(J = t|G = t)P(G = t|B = t, I = t, L = f)P(I = t|B = t, L = f)P(B = t)P(L = f) \\
&= 0.9 \times 0.8 \times 0.5 \times 0.9 \times 0.9 \\
&= 0.2916
\end{aligned}
$$

(c) First, calculate

$$
\begin{aligned}
&P(J = t, B = t, I = t, L = t) \\
&= \sum_g P(J = t, B = t, I = t, L = t, g) \\
&= \sum_g P(J = t|g)P(g|B = t, I = t, L = t)P(I = t|B = t, l = t)P(B = t)P(L = t) \\
&= P(J = t|G = t)P(G = t|B = t, I = t, L = t)P(I = t|B = t, l = t)P(B = t)P(L = t) \\
&+ \\
&P(J = t|G = f)P(G = f|B = t, I = t, L = t)P(I = t|B = t, l = t)P(B = t)P(L = t) \\
&= 0.9 \times 0.9 \times 0.9 \times 0.9 \times 0.1 + 0 \\
&= 0.06561.
\end{aligned}
$$

Then,

$$
\begin{aligned}
&P(J=f,B=t,I=t,L=t)\\
&=\sum_g P(J=f,B=t,I=t,L=t,g)\\
&=\sum_g P(J=f|g)P(g|B=t,I=t,L=t)P(I=t|B=t,l=t)P(B=t)P(L=t)\\
&=P(J=f|G=t)P(G=t|B=t,I=t,L=t)P(I=t|B=t,l=t)P(B=t)P(L=t)\\
&+\\
&P(J=f|G=f)P(G=f|B=t,I=t,L=t)P(I=t|B=t,l=t)P(B=t)P(L=t)\\
&=0.1\times 0.9\times 0.9\times 0.9\times 0.1+1\times 0.1\times 0.9\times 0.9\times 0.1\\
&=0.00729+0.0081=0.01539.
\end{aligned}
$$

And

$$P(J|B=t,I=t,L=t)=\frac{P(J,B=t,I=t,L=t)}{P(B=t,I=t,L=t)}=\alpha P(J,B=t,I=t,L=t),$$

$$0.06561\alpha+0.01539\alpha=1 \implies \alpha\approx 12.34.$$

The probability of this person going to jail is

$$\alpha P(J=t,B=t,I=t,L=t)\approx 0.81.$$

(d) The answer is right if it makes sense. Here provides one possible solution in Figure 2.

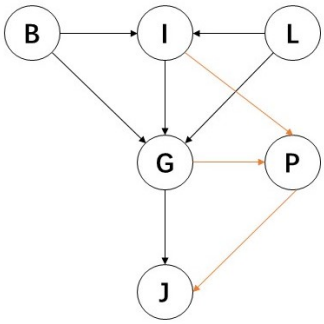


Figure 2: One possible solution

Because a pardon only happens when someone has been indicted and declared guilty. And the presidential pardon can keep he/she out of prison.